ME505 Coding Assignment 1

Opeyemi Akinniyi

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written assignment solutions

(1)

$$u(x) = g_1 - \frac{fx^3}{6E} + \frac{x\left(fL^3 - 6Eg_1 + 6Eg_2\right)}{6EL}$$

(2)

$$u(x) = g_1 - \frac{fx^3}{6E} + \frac{x\left(AfL^2 + 2h\right)}{2AE}$$

(3)

$$u(x) = g_1 + \frac{x\left(fx^3 + \frac{fL^4 - 12Eg_1 + 12Eg_2}{L} - 2Lfx^2\right)}{12E}$$

(4)

$$u(x) = g_1 + \frac{x\left(fx^3 + \frac{2\left(AfL^3 + 6h\right)}{A} - 2Lfx^2\right)}{12E}$$

(i) stiffness matrix for 4 elements:

$$\langle C_2, C_3, C_4\rangle \cdot \frac{EA}{h}\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}\begin{pmatrix} d_2 \\ d_3 \\ d_4 \end{pmatrix} = \langle C_2, C_3, C_4\rangle \cdot \left(\frac{FAh}{2}\begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} + \frac{EA}{h}\begin{pmatrix} u(0) \\ 0 \\ 0 \\ 0 \end{pmatrix} + \frac{EA}{h}\begin{pmatrix} 0 \\ 0 \\ 0 \\ u(L) \end{pmatrix}\right)$$

$u(0) = g_1$ and $u(L) = g_2$

$$\langle C_2, C_3, C_4\rangle \cdot \frac{EA}{h}\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}\begin{pmatrix} d_2 \\ d_3 \\ d_4 \end{pmatrix} = \langle C_2, C_3, C_4\rangle \cdot \left(\frac{FAh}{2}\begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} + \frac{EA}{h}\begin{pmatrix} g_1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \frac{EA}{h}\begin{pmatrix} 0 \\ 0 \\ 0 \\ g_2 \end{pmatrix}\right)$$

$$Kd = F$$

The degree of freedom using symbolic maths on Matlab d is given by:

$$d = \begin{pmatrix} \frac{6Fh^2 + 3Eg_1 + Eg_2}{4E} \\ \frac{4Fh^2 + Eg_1 + Eg_2}{2E} \\ \frac{6Fh^2 + Eg_1 + 3Eg_2}{4E} \end{pmatrix}$$

(ii) stiffness matrix for 3 elements

$$\langle C_2, C_3, C_4\rangle \cdot \frac{EA}{h^e}\begin{pmatrix} -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}\begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix} = \langle C_2, C_3, C_4\rangle \left(\frac{FAh^e}{2}\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ h \end{pmatrix}\right)$$

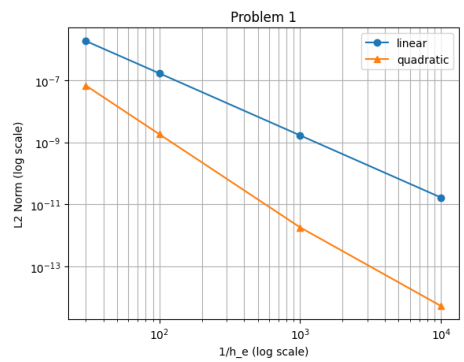

Figure 1: fig: problem 1

if $d_1 = u(0) = g_1$

$$\langle C_2, C_3, C_4 \rangle \cdot \frac{EA}{h^e} \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} d_2 \\ d_3 \\ d_4 \end{pmatrix} = \langle C_2, C_3, C_4 \rangle \left(\frac{FAh^e}{2} \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ h \end{pmatrix} + \frac{EA}{h^e} \begin{pmatrix} g_1 \\ 0 \\ 0 \end{pmatrix} \right)$$

$$Kd = F$$

The degree of freedom using symbolic maths on Matlab d is given by:

$$d = \begin{pmatrix} \frac{2hh_e + 2AEg_1 + 5AFh_e^2}{2AE} \\ \frac{2hh_e + AEg_1 + 4AFh_e^2}{AE} \\ \frac{6hh_e + 2AEg_1 + 9AFh_e^2}{2AE} \end{pmatrix}$$

Coding section

Slopes for Problem 1: linear[-1.99150841 -1.99952544 -1.99996873] Quadratic: [-3.0024891 -3.00000239 -2.53048784]

Slopes for Problem 2: linear[-1.99150841 -1.99948636 -2.00043799] Quadratic [-3.0024891 -3.00000478 -0.80623949]

Slopes for Problem 3: linear[-1.98636918 -1.99907362 -2.00003708] Quadratic: [-2.95961996 -2.99792669 -2.34860547]

Slopes for Problem 4: linear[-1.9863676 -1.99907445 -2.0012469] Quadratic [-2.95961996 -2.99792256 -0.59248482] 1. the problems can be solved exactly if the condition 2n - 1 is met.

2. Optimal convergence is close to 0 on the 1/h$_e axisofthelogscal$

3. using the formular 2n - 1: minimum for problem 1 is 2 for linear, 2 for quad and 3 for cubic. for problem 2 is 2 linear, 2 quadratic and 3 cubic, for problem 3. 2 linear, 3 quadratic and 3 for cubic. for problem 4. 2 linear, 3, quadratic and 3 cubic

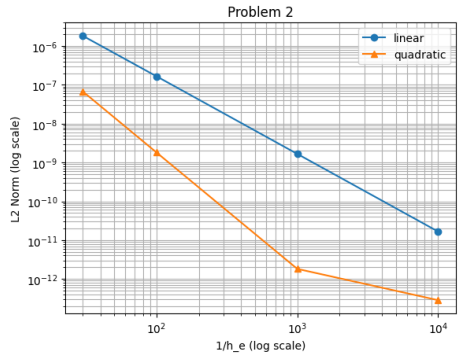

Figure 2: Caption

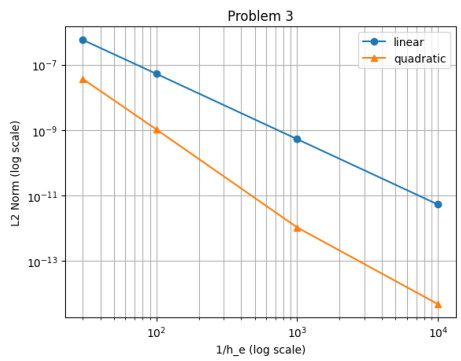

Figure 3: Caption

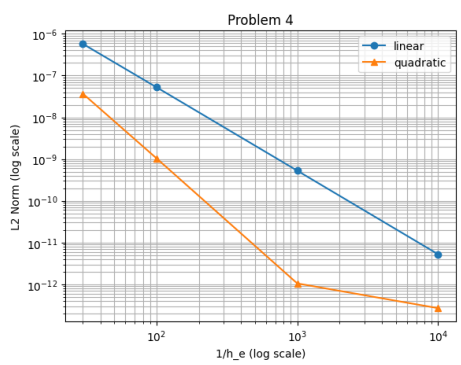

Figure 4: Caption